

The Recursive Observability Filter: A Dynamical-Systems Framework for Civilizational Detectability and the Fermi Paradox

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Abstract

Nearly every quantitative treatment of the Fermi paradox varies how *many* civilizations there are while holding fixed how easy they are to see. We do the opposite. Detectability gets its own equation of motion, and abundance is left alone.

The *Recursive Observability Filter* couples three quantities in dimensionless time — an effective capability I , a regulatory capacity R , and an external observability O — inside the decomposition $N_{\text{obs}} = N_{\text{true}} P_{\text{surv}} h(O) P_{\text{search}}$, which keeps existence, signal production and search coverage from being confused with one another. Two properties hold by construction rather than by clamping, and we prove both: observability can never fall below a thermodynamic floor, and regulatory capacity can never leave the unit interval.

One consequence is the main result. Peak-then-decline observability is the signature that low-observability accounts of the Great Silence normally assume; here it is not assumed. Because the quasi-steady observability is a *ratio* of production to suppression, the decline appears on its own whenever suppression grows faster in capability than production does. Nobody has to put it in by hand.

We also report a failure. The Lambert- W threshold that motivated this work is not a property of the system we integrate: the natural balance cancels I and leaves a logarithm, so the Lambert boundary survives only under a separate balance we have to declare. Nothing here depends on it.

The model is checked against closed-form solutions, an independent high-order integrator, quadrature of the observability equation, and property-based testing over 300 randomised parameter sets, with no violations of either bound. An interactive simulator runs the whole framework in a browser. We do not claim to have solved the Fermi paradox. The contribution is a transparent and falsifiable place to work out what assumptions about recursive intelligence, regulation and observability actually imply — and where they break, to watch them break.

Keywords: Fermi paradox; Drake equation; technosignatures; detectability; recursive intelligence; nonlinear dynamics; Lambert W ; astrobiology; model verification

1 Introduction

1.1 The paradox is an inference problem

The universe is old, planets are everywhere, and a great many planetary systems were finished billions of years before ours got started (Lineweaver, 2001; Borucki, 2016). Nobody has ever confirmed a signal, a probe, an artefact, or an unambiguous technosignature. The tension

between those two statements is what we call the Fermi paradox, and it usually gets compressed into three words: where is everybody? (Hart, 1975; Brin, 1983; Webb, 2015)

Put that way, the question is misleading, because it quietly treats non-observation as evidence about existence. It is not. Non-observation constrains a *product*: how many civilizations arise, how many last long enough to be technologically active, how strong a signature they put out, and whether anyone has yet pointed an instrument at the part of parameter space where that signature would show up (Gray, 2015; Tarter, 2001; Sheikh, 2020). The sharper question is which assumptions would make an extraterrestrial technological civilization visible *to us, now*.

This is not a rhetorical distinction. It decides what a model of the paradox ought to vary. Almost every quantitative treatment, from Drake onward, varies the number of civilizations while holding detectability fixed — usually bundled into a single lifetime parameter L (Prantzos, 2013, 2020). We do the other thing. We hold abundance still and give detectability its own dynamics.

1.2 Existence, detectability, observation

Three quantities get conflated constantly, so we will separate them at the start:

$$N_{\text{true}} \geq N_{\text{detectable}} \geq N_{\text{obs}}, \quad (1)$$

the number of civilizations that exist, the number putting out signatures detectable in principle, and the number our searches have actually found. A civilization can exist without being detectable. It can be detectable in principle and never be observed, simply because nobody looked in the right place. [ESTABLISHED]

This framework modifies the middle term. It asks how a civilization’s external observability changes as its capability, regulatory capacity, energy use, expansion behaviour and information efficiency change — and in particular what happens when it crosses into recursive, self-improving technological capability.

1.3 Why recursive capability is the interesting transition

Something changes in kind when a civilization’s tools start improving the process of making better tools. Automated discovery, engineering and design close a loop between capability and the *rate of growth* of capability (Good, 1965; Bostrom, 2014; Russell, 2019). Quantum computation enters this picture in exactly one way, and it is worth being precise about it: not as a separate phenomenon, but as one possible contributor to how fast that loop can turn, by accelerating the simulation, search and optimisation problems that gate technological progress (Shor, 1994; Nielsen and Chuang, 2010; Preskill, 2018). [SPECULATIVE]

A loop like that plausibly changes more than a growth rate. It may change information efficiency, and so signal leakage. It may change whether expansion looks worth doing. It may stretch or compress the window during which a civilization is loud. None of that is asserted here as fact. The model asks what *follows* if such feedback occurs, and keeps the dependence on that assumption out in the open where it can be attacked.

1.4 What breaks if observability is modelled carelessly

Giving observability its own differential equation is easy to do badly, and the way it fails is instructive enough that we would rather report it than hide it.

Suppose you write the terms representing compression, efficiency and deliberate quietness as an absolute subtraction — a flux removed from O at a rate that does not depend on O . Then nothing whatsoever prevents observability from sailing through zero and running off to large negative values. Worse, a trajectory that has done this still gets classified as “quiet”, because

the classifier only ever asks whether O is below a threshold, and -1431 is certainly below a threshold.

That is not a coding slip. It is a modelling error, and it has a clean physical diagnosis: compression and stealth reduce the signal a civilization *emits*. They are fractional reductions. You cannot suppress emissions that are not being made. Write them as fractional rates acting on the emitted signal, put in an explicit floor for the waste heat nobody can get rid of, and observability becomes bounded below by construction rather than by a clamp bolted on afterwards. We prove this as Proposition 2, with the matching statement for regulatory capacity as Proposition 1.

The correction turns out to carry the main modelling result of the paper. Once suppression is fractional, the quasi-steady observability is a *ratio* of production to suppression — and peak-then-decline behaviour then follows automatically whenever suppression grows faster in capability than production does. It no longer has to be imposed by picking a peaked functional form. That matters, because a low-observability account of the Great Silence that assumes a peaked $O(I)$ has assumed its conclusion. One that derives the peak from two independently motivated mechanisms has not.

1.5 Contributions

1. An observability-aware decomposition $N_{\text{obs}} = N_{\text{true}} P_{\text{surv}} h(O) P_{\text{search}}$ that separates survival, signal production and search coverage, and says exactly where a dynamical detectability model acts on the Drake equation (§3).
2. A three-variable dynamical model of capability, regulation and observability, built from mechanism balances, with two structural guarantees — $O \geq O_{\text{floor}}$ and $R \in [0, 1]$ — that hold by construction and not by clamping (§4).
3. Peak-then-decline observability as a *derived* consequence of suppression outgrowing production, rather than an assumed functional form (§7).
4. A registry architecture in which every functional form is interchangeable, so that “do the conclusions survive changing the functions?” becomes a menu selection instead of a rewrite (§4).
5. A verification protocol that is unusual in this literature: closed-form solutions, an independent high-order integrator, quadrature, and property-based testing over randomised parameter sets (§6).
6. An explicitly negative result about the Lambert- W threshold that started this work: the balance justifying W is not the balance governing the system we integrate (§5).

1.6 What this paper does not claim

It does not claim that extraterrestrial civilizations exist, that advanced civilizations go quiet, that recursive artificial intelligence causes collapse or transcendence, or that the Fermi paradox has been solved. Every substantive statement in what follows carries a strength label — [ESTABLISHED], [PLAUSIBLE], [HYPOTHETICAL] or [SPECULATIVE] — and they are all collected in Table 6 so that nobody has to hunt for the weak ones.

2 Background

2.1 The Drake equation and what exoplanet science settled

The Drake equation organises the expected number of communicative civilizations as a product,

$$N = R_* f_p n_e f_l f_i f_c L, \tag{2}$$

and it is best read as a map of our ignorance rather than as a calculator (Drake and Sobel, 1992). Exoplanet surveys have since pinned down its leading astrophysical factors rather well: planets are common, and potentially habitable planets are no longer speculative (Borucki, 2016; Batalha, 2014; Dressing and Charbonneau, 2015; Kopparapu et al., 2013; Seager, 2013). [ESTABLISHED]

So the uncertainty has migrated rightward. It now sits in the emergence of life and intelligence, and above all in L , the interval over which a civilization is detectable. Prantzos (2020) shows explicitly that L dominates the whole inference. That is the term this framework acts on, by refusing to let it be a scalar.

2.2 Great Filter and existential risk

The Great Filter frames the problem as a bottleneck: somewhere between non-living matter and a long-lived expansive civilization there must be a step that is extraordinarily improbable (Hanson, 1998). Whether that step is behind us or ahead of us carries obvious stakes, and the existential-risk literature explores the second possibility at length (Bostrom, 2002, 2013; Ord, 2020; Häggström, 2016). Ćirković (2018) surveys the philosophical terrain and Verendel and Häggström (2017) give a Bayesian treatment.

But the Great Filter is a statement about where probability mass goes, not about dynamics. It says nothing about how a civilization moves through the bottleneck, and nothing at all about what it looks like from outside while it is doing so. That is the gap a dynamical model can fill.

2.3 Technosignatures and the search

What could actually be detected has been studied since Cocconi and Morrison (1959) and Dyson (1960). Energy-use signatures scale with the Kardashev classification (Kardashev, 1964); infrared searches for large-scale energy capture have returned null results at survey scale (Wright et al., 2014; Griffith et al., 2015; Garrett, 2015); and the field has increasingly formalised how to reason about coverage of the search space (Tarter, 2001; Wright, 2018; Sheikh, 2020; Wright et al., 2022).

Two points matter for what follows. Different signature classes correspond to different civilizational behaviours — waste heat, narrowband beacons, propulsion, megastructures, atmospheric markers — so “detectability” is not one quantity, and treating it as one throws away most of the information. And the searches conducted so far cover a genuinely small fraction of the available parameter space, which is exactly why P_{search} has to appear explicitly in any honest inference (Wright et al., 2022). [ESTABLISHED]

2.4 Expansion, percolation and timing

If interstellar settlement is feasible at all, the galaxy should be crossed quickly on astronomical timescales (Hart, 1975; Tipler, 1980; Armstrong and Sandberg, 2013), which sharpens the paradox considerably. Percolation and agent-based treatments soften it again by letting settlement stall (Landis, 1998; Forgan, 2009). Anthropic and hard-steps arguments address why we find ourselves so early (Carter, 1983; Watson, 2008; Kipping, 2020).

Then there is the grabby-aliens analysis of Hanson et al. (2021), which makes a sharp observation: if loud expanding civilizations are what explain human earliness, then quiet ones have to be rare as well. That is the most direct challenge to any low-observability account of the Great Silence, ours very much included, and we return to it in §8.

2.5 Recursive intelligence

The possibility that capability improves the machinery of capability has been discussed since Good (1965), and elaborated with attention to control and alignment by Bostrom (2014), Russell (2019), Omohundro (2008) and Yudkowsky (2008). Quantum computation is relevant to this

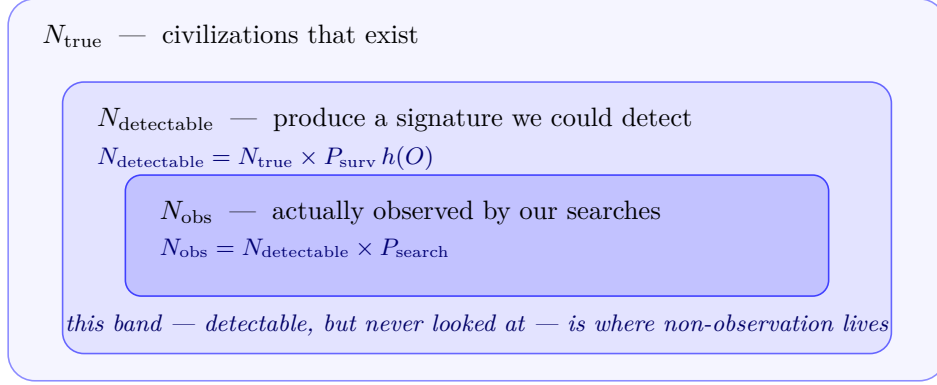


Figure 1: Existence, detectability and observation are three different things. Each band is reached from the one outside it by the factor written beneath its name. A null result constrains only the innermost set, and therefore only the *product* in Eq. (3) — never N_{true} on its own.

framework through that channel and no other — as one possible accelerant of the recursive improvement loop (Shor, 1994; Preskill, 2018; Nielsen and Chuang, 2010) — and we are explicit that both its magnitude and its sign of effect are unconstrained. [SPECULATIVE]

2.6 The gap

Across all of these literatures, detectability is treated as a static property: a coefficient, a lifetime, or a survey limit. Where it is allowed to change, the change is imposed as a chosen profile. What is missing is a formulation in which observability has its own equation of motion, is coupled to the same processes that drive capability and regulation, and is constrained to stay physical. Supplying that, and then testing it seriously, is what this paper is for.

3 Separating existence, detectability and observation

3.1 The decomposition

Let N_{true} be the number of technological civilizations that exist in the volume and epoch we care about. Write P_{surv} for the chance that such a civilization is still around in the phase we could detect at all, $h(O)$ for the chance it is putting out a signature strong enough to register given its external observability O , and P_{search} for the chance our searches have actually looked where that signature is. Then

$$N_{\text{obs}} = N_{\text{true}} \cdot P_{\text{surv}} \cdot h(O) \cdot P_{\text{search}} \quad (3)$$

with h a saturating map from signature strength to detection probability. We take

$$h(O) = 1 - e^{-\kappa O}, \quad (4)$$

where κ sets instrument sensitivity, and we write $P_{\text{det}} = h(O) P_{\text{search}}$ for the detection probability of a single civilization.

There is no theory in Eq. (3). It is bookkeeping. Its whole value is that it refuses to let four different mechanisms hide inside one number. [ESTABLISHED] Figure 1 shows how the sets nest, and Figure 2 shows where this framework acts on the Drake equation.

3.2 Why this matters for inference

Suppose a survey reports nothing. Under Eq. (3) the likelihood of that outcome depends on P_{search} and on how O is spread across the population — not on N_{true} by itself. Two very different

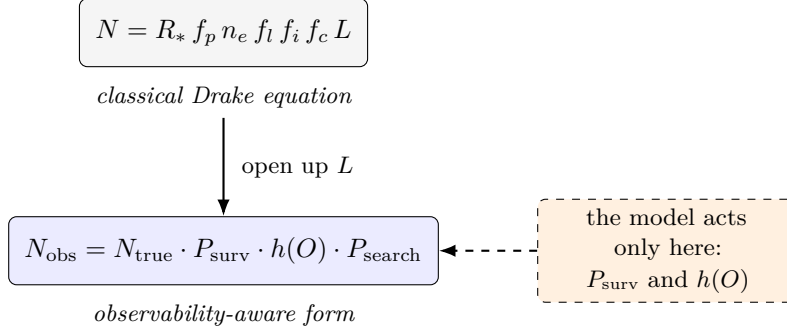


Figure 2: This framework refines the Drake equation rather than replacing it. The astrophysical factors are left exactly as they are; all the new machinery sits inside survival and signal production.

universes give the same null result: one where civilizations are rare, and one where they are everywhere but mostly quiet during the sliver of time we can sample. Telling them apart needs a model of how O is distributed and how it changes, which is what §4 supplies. [ESTABLISHED]

3.3 What observability is, and is not

By O we mean *external observability*: the strength of whatever leaves the system, adding up waste heat, leakage, deliberate emission, expansion footprint and engineered structures. It is not energy use and it is not capability. A civilization can be enormously capable and rather quiet; it can also be modest and extremely loud.

And O can get small, but it cannot get to zero. Using energy has thermodynamic consequences, and claiming otherwise would need physics we have no right to assume (Dyson, 1960; Kardashev, 1964). So we build that in: O is bounded below by a strictly positive floor O_{floor} (Proposition 2). In this framework low observability always means hard to find. It never means gone. [ESTABLISHED]

4 The dynamical model

4.1 Variables

Three quantities evolve together in a dimensionless time τ .

$I(\tau)$ — **effective capability**. A compressed systems variable that rolls together scientific capability, automation, computational capacity, coordination and technological leverage. It is not a measure of intelligence in the psychometric sense, not a measure of moral worth, and not a measure of consciousness. A civilization can grow enormously in I without becoming wiser, safer or louder.

$R(\tau)$ — **regulatory capacity**. Alignment, governance, safety culture, institutional learning, epistemic reliability, error correction: whatever it takes to stop a runaway. We treat it as a *quality index* normalised to $[0, 1]$, and that normalisation is what makes it commensurable with the growth rates it has to oppose (§4.3).

$O(\tau)$ — **external observability**. Signature strength as defined in §3, bounded below by $O_{\text{floor}} > 0$.

Table 1 lists these with their drivers. Figure 3 maps the usual civilizational stages onto them, and Figure 4 shows what is coupled to what.

Table 1: Model variables and drivers. The three states are integrated; the drivers are exogenous inputs the user sets.

Symbol	Role	Interpretation
I	state	effective capability: science, automation, computation, coordination
R	state	regulatory quality index in $[0, 1]$: alignment, governance, error correction
O	state	external observability: leakage, waste heat, expansion footprint
A	driver	ordinary amplification pressure (baseline technological leverage)
A_{rec}	driver	recursive amplification: self-improvement, automated design, compute scaling
A_{ref}	driver	reflective amplification: capability aimed at safety and foresight
Q	driver	institutional quality

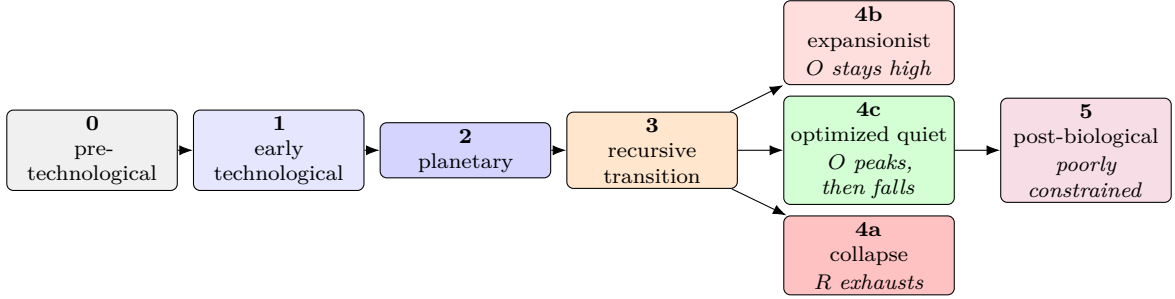


Figure 3: Civilizational stages, and the branch at the recursive transition. Stages 0–2 are pre-recursive and behave the way anyone would expect. Everything interesting happens at stage 3, where the same machinery can go three different ways depending on which term wins.

4.2 Capability

Capability grows two ways: by ordinary accumulation, aAI , and by recursive amplification, $bA_{\text{rec}}F(I)$, where F says how aggressive the recursion is. It is held back by regulatory friction, cRI , and it eventually runs into physical, energetic and coordination limits, $s_I I^2$:

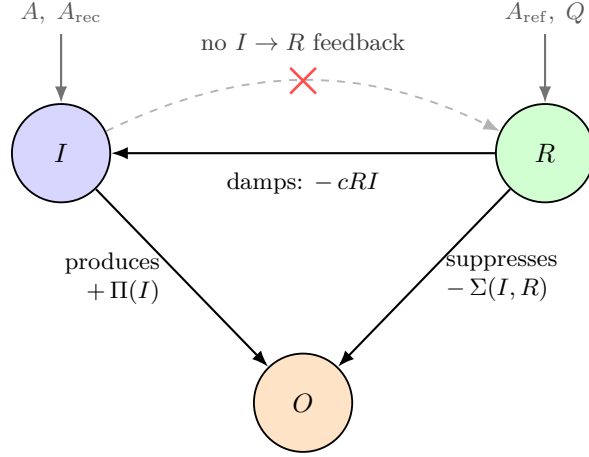
$$\frac{dI}{d\tau} = aAI + bA_{\text{rec}}F(I) - cRI - s_I I^2. \quad (5)$$

Every term is there because some mechanism puts it there. That is the point: the equation is assembled out of balances you can argue with, not written down because it integrates nicely.

4.3 Regulation

Regulatory capacity is built by reflective capability and institutional quality, and eroded by the sheer pace of recursive amplification. The modelling choice that matters is this: erosion has to be proportional to the regulation that is actually there, because you cannot erode capacity that does not exist, and building has to act on whatever headroom is left. So

$$\frac{dR}{d\tau} = \underbrace{(uA_{\text{ref}} + vQ)}_B (1 - R) - \underbrace{(wA_{\text{rec}} + s_R)}_W R. \quad (6)$$



$$h(O) P_{\text{search}} \longrightarrow N_{\text{obs}}$$

Figure 4: What is coupled to what. Regulation damps capability through $-cRI$. Observability is produced by capability and suppressed fractionally by capability *and* regulation acting together — Σ depends on both, even though the arrow has to be drawn from one of them. The struck-out arc is the coupling this model does *not* have: in Eq. (6), regulatory capacity responds to the exogenous drivers and to erosion by A_{rec} , but never to capability itself. Closing that loop is the most interesting extension we can see (§9).

Proposition 1 (Regulation is confined to the unit interval). *If $B \geq 0$ and $W \geq 0$ and $R(0) \in [0, 1]$, then $R(\tau) \in [0, 1]$ for all $\tau > 0$, and*

$$R_{\infty} = \frac{B}{B + W} = \frac{uA_{\text{ref}} + vQ}{uA_{\text{ref}} + vQ + wA_{\text{rec}} + s_R}. \quad (7)$$

Proof. At $R = 0$, Eq. (6) gives $dR/d\tau = B \geq 0$, so the lower boundary cannot be crossed downwards. At $R = 1$ it gives $dR/d\tau = -W \leq 0$, so the upper boundary cannot be crossed upwards. With constant drivers Eq. (6) is linear in R , $dR/d\tau = B - (B + W)R$, which gives the stated fixed point and $R(\tau) = R_{\infty} + (R_0 - R_{\infty})e^{-(B+W)\tau}$. $\square$

Two things follow, and both are worth pausing on. First, R_{∞} is a *share*: the fraction of the total pressure on governance that happens to be constructive. That makes it something you can reason about directly. Second, because R can never go negative, the damping term $-cRI$ in Eq. (5) can never flip sign. Let R go negative and failing governance starts *accelerating* capability, which is not a mechanism anybody means to model — it is an arithmetic accident wearing a mechanism’s clothes.

4.4 Observability

Observability is produced by energy use, broadcast leakage and expansion footprint, and suppressed by compression and stealth. The suppression is fractional: it acts on the signal actually being emitted, above an irreducible floor.

$$\frac{dO}{d\tau} = \underbrace{pE(I) + qB_c(I) + rX(I)}_{\Pi(I)} - (O - O_{\text{floor}}) \underbrace{(mC(I, R) + nS(I, R))}_{\Sigma(I, R)}. \quad (8)$$

Proposition 2 (Observability is bounded below by the floor). *If $\Pi \geq 0$ and $\Sigma \geq 0$ and $O(0) \geq O_{\text{floor}}$, then $O(\tau) \geq O_{\text{floor}}$ for all $\tau > 0$. Where $\Sigma > 0$, the quasi-steady observability is*

$$O^* = O_{\text{floor}} + \frac{\Pi(I)}{\Sigma(I, R)}. \quad (9)$$

Table 2: Interchangeable functional forms. Any combination is admissible; the solver never refers to a specific choice.

Component	Variants	Form
$F(I)$ recursion	linear, superlinear, Lambert, exponential, saturating	$I; I^2; Ie^I; e^I; I/(1 + I/K)$
$O(I)$ static	A increasing, B decreasing, C peaked, D threshold	$1 - e^{-\lambda I}; e^{-\lambda I}; Ie^{-\lambda I}; \sigma(k(I - I_c))$
E energy use	linear, Kardashev	$I; I^2$
B_c broadcast	linear, decaying	$I; Ie^{-\lambda I}$
X expansion	linear, inward	$I; 0$
C, S suppression	linear, superlinear	$IR; I^2R$

Proof. At $O = O_{\text{floor}}$ the suppression term vanishes identically, leaving $dO/d\tau = \Pi \geq 0$, so the floor cannot be crossed downwards. Setting $dO/d\tau = 0$ in Eq. (8) and solving for O gives Eq. (9). $\square$

Equation (9) is the analytical heart of this paper, and the reason is that O^* is a *ratio*. What it does as capability grows is decided by how production and suppression scale *against each other*, never by either one alone:

$$\Pi \sim I, \quad \Sigma \sim IR \implies O^* \sim \frac{1}{R} \quad (\text{observability independent of capability}), \quad (10)$$

$$\Pi \sim I, \quad \Sigma \sim I^2R \implies O^* \sim \frac{1}{IR} \quad (\text{observability declines with capability}). \quad (11)$$

Look at what the second line does. It produces peak-then-decline without anyone choosing a peaked function — the shape falls out of the ratio. That is what we mean by the signature being derived rather than assumed, and §7 shows it happening numerically. [PLAUSIBLE]

4.5 Interchangeable functional forms

None of the functional forms above is privileged, and we would rather not pretend otherwise. Each is drawn from a registry (Table 2), so asking whether a conclusion survives a change of functions is a menu selection instead of a rewrite. That is what turns robustness from an aspiration into something you can actually check in an afternoon.

4.6 Regime classification

Trajectories are labelled by the criteria in Table 3, checked in a fixed order. The order is not commutative — a trajectory that meets two criteria gets the earlier label — and we say so explicitly because it changes how the phase portrait in §7 should be read.

Regime 1 is called a *proxy* on purpose. There is no mortality term anywhere in this model, so what it registers is the conditions under which collapse should be expected — not a collapse. We come back to this in §9, because it is a real limitation and not a quibble.

5 The Lambert threshold, and why it is auxiliary

This section reports a negative result about our own starting point.

The Lambert W function motivated the original formulation of this work, and it is mathematically correct everywhere we use it. But the balance that justifies W turns out not to be the balance governing the system we actually integrate. We could have quietly kept the boundary

Table 3: Regime criteria, in evaluation order. Thresholds are model parameters, not empirical quantities.

#	Regime	Criterion	Strength
1	collapse proxy	runaway <i>and</i> ($R < R_{\min}$ or O peaked then fell below $O_{\text{detectable}}$)	[HYPOTHETICAL]
2	runaway	$I > I_{\text{runaway}}$ or $A_{\text{rec}}/(R + \epsilon) > \Theta$	[SPECULATIVE]
3	pre-detectable	$\max O < O_{\text{detectable}}$ and $\max I < I_{\text{advanced}}$	[PLAUSIBLE]
4	optimized low-observable	$I \geq I_{\text{advanced}}, R \geq R_{\min}, O < O_{\text{detectable}}$	[HYPOTHETICAL]
5	expansionist visible	$I \geq I_{\text{advanced}}$ and $O \geq O_{\text{detectable}}$	[HYPOTHETICAL]
6	visible technological	$P_{\text{det}} > 0.01$	[PLAUSIBLE]
7	uncertain	none of the above	[PLAUSIBLE]

and let it look like a property of the dynamics. We would rather set the whole thing out, because noticing this kind of gap is exactly what separates a model you can use from mathematics you can admire.

5.1 When W is actually required

The Lambert W function is defined implicitly by $W(y)e^{W(y)} = y$, and you need it precisely when an unknown shows up both inside and outside an exponential (Corless et al., 1996). A relation

$$xe^x = y \implies x = W(y) \quad (12)$$

genuinely requires it. A relation $e^x = y$ does not; that is a logarithm. Any appearance of W that does not rest on the first structure is decoration.

5.2 The natural balance does not need W

Take the Lambert recursion kernel $F(I) = Ie^I$ from Table 2. Equation (5) becomes

$$\frac{dI}{d\tau} = aAI + bA_{\text{rec}}Ie^I - cRI - s_I I^2. \quad (13)$$

Set recursive amplification against regulatory damping, drop ordinary growth and saturation for the moment, and you get

$$bA_{\text{rec}}Ie^I = cRI. \quad (14)$$

Now look at what happens. For $I > 0$ the factor I *cancels on both sides*, and we are left with

$$bA_{\text{rec}}e^I = cR \implies I_{\text{thr}} = \ln\left(\frac{cR}{bA_{\text{rec}}}\right), \quad (15)$$

a *logarithmic* threshold. The Lambert structure is destroyed by the very cancellation that the Ie^I kernel appeared to guarantee. The damping term carries a factor of I too, and the two factors annihilate. This is the central observation of the section, and it is remarkably easy to miss — we missed it for a while.

5.3 Recovering Lambert structure requires a different balance

Lambert structure survives only if Ie^I is set against a regulatory capacity that is *not* itself proportional to I . Introduce such a capacity,

$$C_R = cR^\eta \ln(1 + A_{\text{ref}}) \mathcal{E}, \quad (16)$$

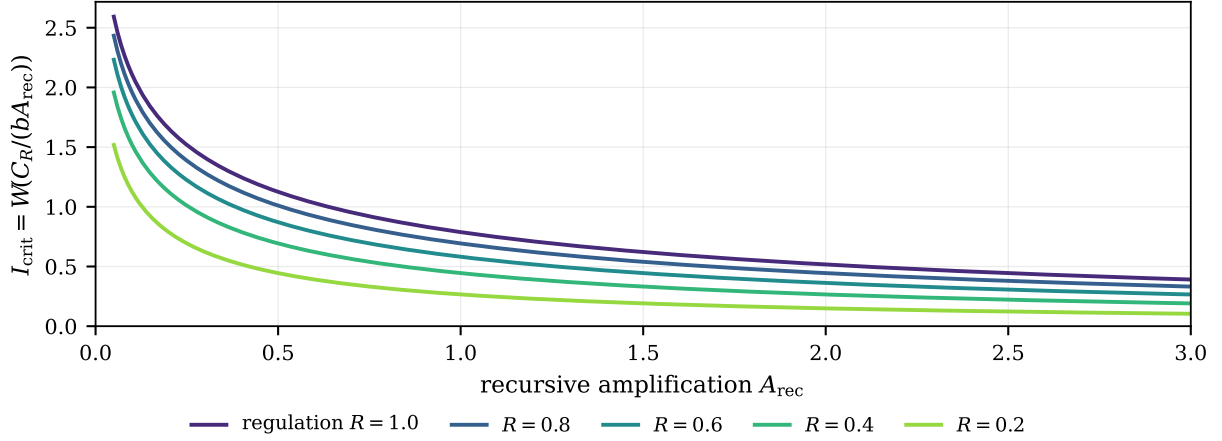


Figure 5: The auxiliary Lambert boundary I_{crit} against recursive amplification, at five fixed values of regulatory capacity, from Eq. (17). More regulation raises the critical capability that can be absorbed; stronger recursive amplification lowers it, and does so steeply at small A_{rec} . This is a diagnostic evaluated under the auxiliary balance of Eq. (16), not a bifurcation set of the integrated system.

with η a regulation exponent and $\mathcal{E}$ an alignment factor, impose $bA_{\text{rec}}Ie^I = C_R$, and now nothing cancels:

$$Ie^I = \frac{C_R}{bA_{\text{rec}}} \quad \Rightarrow \quad \boxed{I_{\text{crit}} = W\left(\frac{c R^\eta \ln(1 + A_{\text{ref}}) \mathcal{E}}{bA_{\text{rec}}}\right)}. \quad (17)$$

Equation (17) is a clean Lambert boundary, and it is what Figure 5 shows.

5.4 The honest reading

Here is the difficulty. C_R appears nowhere in Eq. (5). The system we integrate contains the damping term $-cRI$, whose threshold is Eq. (15) — logarithmic. So Eq. (17) describes a balance we have *posited alongside* the dynamics, not one the dynamics contain. Stating it plainly:

The Lambert boundary $I_{\text{crit}} = W(C_R/bA_{\text{rec}})$ is an *auxiliary stability diagnostic* evaluated under a stated auxiliary balance. It is not a bifurcation of the integrated system, and no result in §7 depends on it. [HYPOTHETICAL]

The implementation enforces that distinction rather than trusting the prose to carry it: the Lambert branch is guarded so W is evaluated only when the Lambert kernel is genuinely selected, and every other kernel falls back to the logarithmic threshold of Eq. (15). Verification of the identity $I_{\text{crit}}e^{I_{\text{crit}}} = C_R/(bA_{\text{rec}})$ to machine precision is reported in §6.

5.5 Why keep it at all

Three reasons. It gives an analytic handle on the intuition that regulation can only absorb so much recursive amplification. It makes the comparison between regulatory capacity and amplification pressure explicit and plottable. And, reported honestly, it is a worked example of the discipline the rest of the framework demands: a candidate piece of mathematics, tested to see whether it was really required, and found to be required only under an assumption we then have to declare out loud.

6 The simulator, and how we checked it

Models like this one are usually verified about as far as “the code ran and the plots looked reasonable”. That is not enough here. Every conclusion in §7 is a statement about the *shape*

of a computed trajectory, so if the trajectory is wrong the conclusion is wrong, and it will be wrong in a way that still looks perfectly plausible on a screen. We therefore treat verification as part of the contribution rather than as housekeeping.

6.1 The simulator

The framework is packaged as an interactive simulator. Every driver, weight and threshold in Appendix C is exposed as a control; every entry in the registry of Table 2 is a menu selection; and the eight stage presets of §7 load with one click. Trajectories, the phase portrait, the regime label and the detection probability P_{det} all update as the parameters move.

We mention this because it is the part of the work that is hardest to convey on paper. A reader who suspects that some result here is an artefact of a particular parameter choice can go and drag the slider. The registry means that “does this conclusion survive a different functional form?” takes a few seconds to answer instead of an afternoon of editing, and we would rather people attacked the model that way than took our word for it. The simulator runs in a browser with nothing to install; the address is given at the end of the paper.

6.2 Integration

The system (5)–(8) is integrated with an explicit adaptive Dormand–Prince Runge–Kutta method of order 5(4) with dense output (Dormand and Prince, 1980; Virtanen et al., 2020; Harris et al., 2020), at relative tolerance 10^{-6} and absolute tolerance 10^{-9} . All exponentials are argument-clipped to $[-50, 50]$ to stop them overflowing.

Two of the recursion kernels in Table 2 — superlinear and exponential — blow up in finite time: I genuinely diverges at a finite τ . An adaptive stepper walking into a singularity takes smaller and smaller steps and never arrives, so integration stops on a terminal event when I exceeds a bound I_{abort} set two orders of magnitude above the runaway threshold, and the remaining output samples hold the terminal state.

That last detail is worth a sentence, because we got it wrong first. If you pad instead with the last *output* sample, the recorded maximum can understate the divergence by orders of magnitude — the singularity almost always falls between two samples, and the trajectory looks merely large rather than infinite.

6.3 Five independent oracles

Each check below tests the implementation against something that is not the implementation. Results are collected in Table 4.

A. Closed-form regulation. Equation (6) is linear in R and — this is the useful part — completely independent of I and O . So it integrates exactly, giving $R(\tau) = R_{\infty} + (R_0 - R_{\infty})e^{-(B+W)\tau}$ with R_{∞} from Eq. (7). Every configuration reproduces it to $\leq 2.3 \times 10^{-6}$, which is what the requested tolerance should give.

B. An independent high-order integrator. Every configuration was integrated again with an eighth-order Dormand–Prince method at tolerances $10^{-12}/10^{-14}$ and compared against the production solver at its default settings. All twenty agree. For the two blow-up kernels the comparison stops at the abort event, since the reference integrator has no such guard and simply keeps going.

C. Quadrature of the observability equation. $dO/d\tau$ was rebuilt from the computed I and R and integrated by quadrature, which never touches the ODE solver’s own treatment of O . Agreement is $\leq 1.5 \times 10^{-4}$ on every non-singular configuration.

Table 4: Verification results. Each oracle tests the implementation against something other than itself.

	Oracle	Scope	Result
A	closed-form $R(\tau)$	19 configurations	max error 2.3×10^{-6}
B	independent 8th-order integrator	20 configurations	20/20 agree
C	quadrature of $dO/d\tau$	12 non-singular configurations	$\leq 1.5 \times 10^{-4}$
D	Lambert identity	2304 evaluations	4.4×10^{-16}
E	structural invariants	300 random configurations	0 violations

D. The Lambert identity. I_{crit} has to satisfy its own definition, $I_{\text{crit}}e^{I_{\text{crit}}} = C_R/(bA_{\text{rec}})$. Across 576 parameter combinations and four values of R — 2304 evaluations — the largest relative residual is 4.4×10^{-16} , which is machine precision. The validity guard behaves as specified too, returning no boundary at all for any non-Lambert kernel.

E. Property-based testing of the invariants. This is the one that matters most. Propositions 1 and 2 claim *structural* bounds, so they have to hold for arbitrary admissible parameters — not for the eight presets we happened to tune. Three hundred randomised parameter sets were drawn across the full declared ranges, with randomised recursion kernels and randomised suppression forms. There were **no violations**: the global minimum observability came out at 2.16×10^{-3} , comfortably above the floor, and regulation stayed inside $[4.0 \times 10^{-3}, 0.999]$.

A bound that held only for the configurations we chose would not be a bound. It would be a coincidence we had grown fond of. This test is the difference.

6.4 What this establishes, and what it does not

These checks establish that the equations we have written down are the equations being solved, and that they are being solved accurately. They establish exactly nothing about whether those equations describe a real civilization. Not one term here has been calibrated against data, for the excellent reason that no relevant data exist.

The distinction between a model that is *verified* and one that is *validated* is worth keeping sharp, and it gets blurred constantly in this subject. We claim the first and make no claim at all to the second. [ESTABLISHED]

7 Results

Everything below uses the parameter set in Appendix C, and every number can be reproduced from the simulator.

7.1 Stage trajectories

Figure 6 shows capability, regulation and observability for the eight canonical stage configurations. Table 5 sets what each stage actually did against what that stage is *defined* to do. This is a low bar, and we report it because it is a bar models fail all the time without anyone noticing — a stage called “collapse” that quietly grows is the sort of thing that survives for years inside a plausible-looking plot.

Two configurations reward a closer look. Stage 1, the early technological phase, has no regulatory damping at all, so capability reduces to a plain logistic equation and hits its analytic limits exactly: $I \rightarrow aA/s_I = 3.0$ and $O \rightarrow 1 - e^{-\lambda I} = 0.5934$ (Appendix B). Stage 4a drives regulation down to $R_\infty = 0.0438$, below the collapse threshold $R_{\text{min}} = 0.05$ — and it gets there *without R ever going negative*. Collapse happens by regulatory exhaustion, the mechanism we meant to model, rather than by a sign flip in the damping term, which is the mechanism we did not.

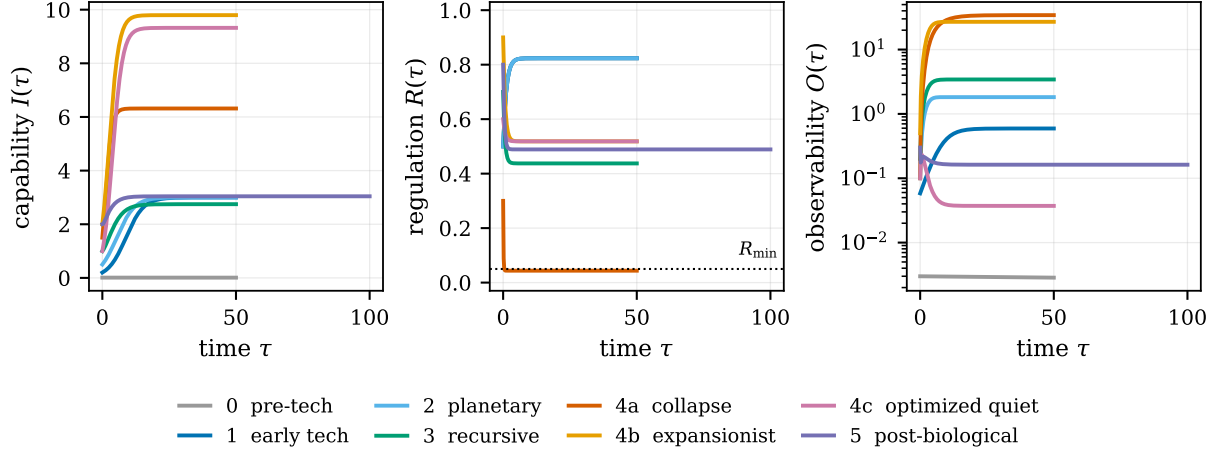


Figure 6: Capability, regulation and observability across the eight stage configurations. Regulation stays inside $[0, 1]$ by Proposition 1. Observability is plotted logarithmically and stays strictly positive by Proposition 2, spanning nearly five orders of magnitude from the quietest stage to the loudest.

Table 5: What each stage was specified to do, against what it did.

Stage	Specified behaviour	Simulated	Regime
0	$dI/d\tau \approx 0$, no signature	$I = 0.010$, $O = 0.003$	pre-detectable
1	$I \rightarrow aA/s_I$, O rising	$I = 3.0000$, $O = 0.5934$	visible technological
2	capability grows, O still rising	$I = 3.000$, $O : 0.10 \rightarrow 1.82$	visible technological
3	full model, recursion active	$I = 2.75$, $O = 3.43$	visible technological
4a	regulation collapses	$R \rightarrow 0.0438 < R_{\min}$, $R \geq 0$	collapse proxy
4b	high I , high O sustained	$I = 9.80$, $O = 27.0$	expansionist visible
4c	O peaks, then declines	$0.100 \rightarrow 0.231 \rightarrow 0.037$	optimized low-observable
5	speculative	$I = 3.04$, $O : 0.30 \rightarrow 0.16$	visible technological

7.2 Peak-then-decline is derived, not imposed

This is the central result, and Figure 7 is the whole argument in one picture. Both panels are the *same* stage-4c configuration. Only the form of the suppression term differs.

Panel (a) uses an absolute suppression term, $-mC - nS$, whose rate does not depend on O at all. Observability passes through zero at $\tau \approx 0.3$ and then just keeps going, reaching -1431 by the end of the run. And here is the part that should bother you: under the criteria of Table 3 that trajectory is still labelled “optimized low-observable”, because the classifier looks at $O < O_{\text{detectable}}$ and finds it satisfied. The label is arithmetically correct and physically empty. The civilization is not quiet. It has fallen off the bottom of the model.

Panel (b) uses the fractional form of Eq. (8). Observability rises from 0.100 to a peak of 0.231 at $\tau = 0.65$, then decays to 0.037: below the detection threshold, above the thermodynamic floor, and strictly positive the entire time.

Now, why does it decline? Nothing in this configuration picks a peaked functional form — O is integrated, not evaluated from some curve we chose in advance. The decline comes straight out of Eq. (11). Compression and stealth scale as $I^2 R$ while production scales as I , so the quasi-steady observability $O^* \sim 1/(IR)$ has to fall as capability grows. The civilization gets harder to see because its efficiency outruns its output. That is a mechanism, not a stipulation. [\[PLAUSIBLE\]](#)

The distinction is the whole answer to the standing objection against low-observability accounts of the Great Silence — that they assume the very thing they set out to explain. A model that selects a peaked $O(I)$ does assume it, and the objection lands. A model that gets the peak out of the ratio of two independently motivated terms does not, and has to be attacked

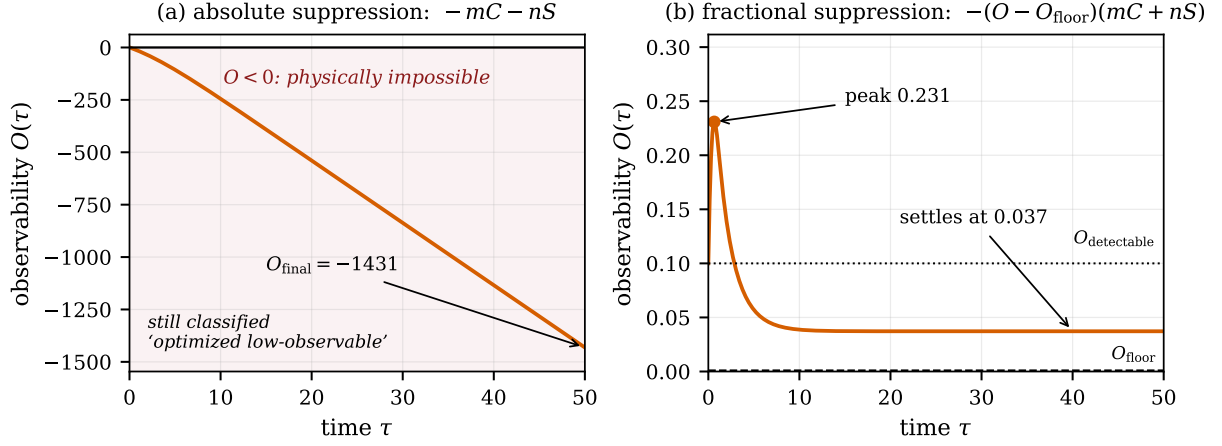


Figure 7: Why the form of the suppression term decides everything. Both panels are the same configuration. (a) With absolute suppression the trajectory passes through zero and dives, spending the entire run inside the shaded region the model’s own thermodynamics forbid — and it is *still* classified “optimized low-observable”. (b) With fractional suppression and a thermodynamic floor, the same configuration gives a genuine peak-then-decline, bounded below. The decline is derived from suppression outscaling production, not imposed by choosing a peaked profile.

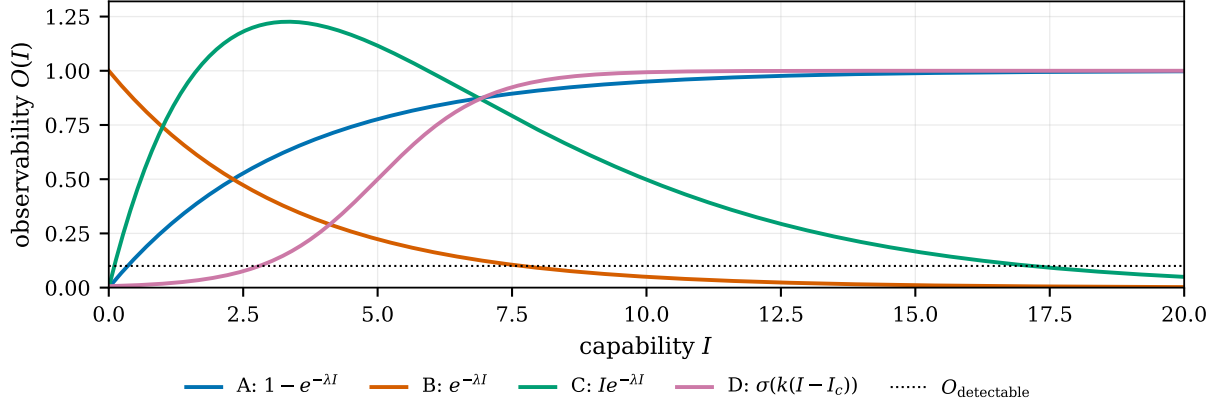


Figure 8: The four interchangeable static observability models: A rises and saturates, B falls from the start, C rises then falls, D switches on at a threshold. Any conclusion about the Great Silence has to be reported together with the model that produced it — the low-observability regime does not arise under model A at all.

somewhere else: on whether the required scaling separation is physically plausible. We think that is a much better argument to be having.

7.3 Robustness across observability models

Figure 8 shows the four static observability models from Table 2: one rising, one falling, one peaked, one switching on at a threshold. The framework admits all four.

The useful robustness question is not which curve is right. Nobody knows which curve is right. The question is whether a conclusion survives changing it. The low-observability regime is reachable under the peaked and decreasing models and under the dynamical form of Eq. (8). It is *not* reachable under the increasing model, which instead gives the expansionist-visible regime we see in stage 4b.

We count that as the framework working correctly. A model in which every functional choice led to the same conclusion would not be testing anything at all; it would be a machine for producing the answer we already had. [ESTABLISHED]

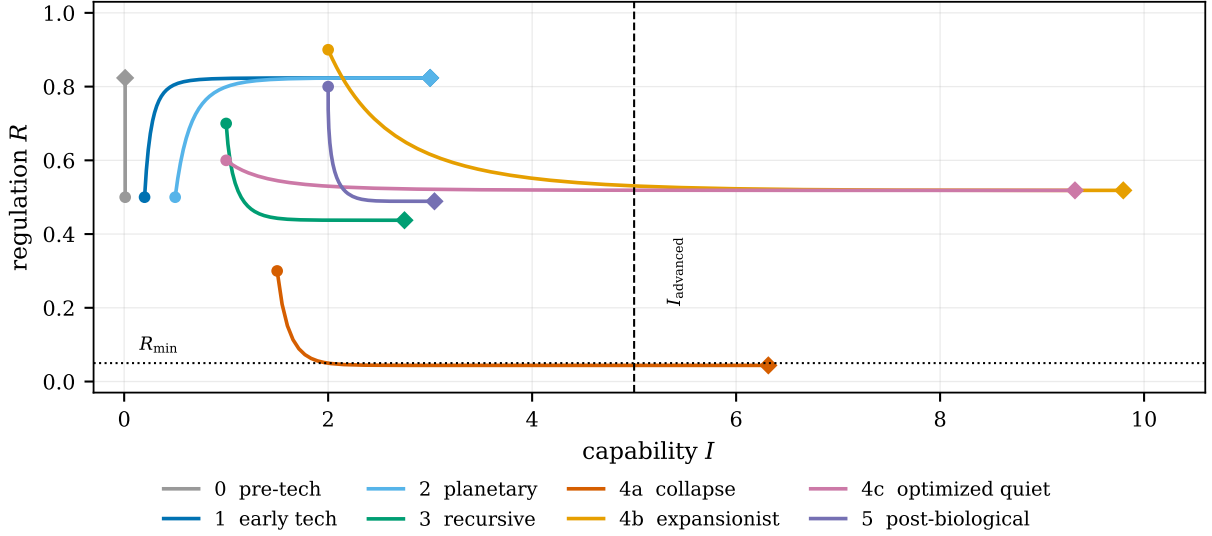


Figure 9: Phase portrait in the capability–regulation plane. Circles mark initial conditions, diamonds mark final states. Regulation is bounded to $[0, 1]$, so the reachable region is a strip, and the collapse boundary $R_{\min}$ is approached but never crossed into negative values.

7.4 Phase structure

Figure 9 puts the trajectories in the (I, R) plane. Because regulation is confined to $[0, 1]$, the reachable phase space is a bounded strip rather than a half-plane, and that makes the regime boundaries directly readable off the picture: the vertical line at I_{advanced} separates pre-advanced from advanced capability, and the horizontal line at $R_{\min}$ marks regulatory exhaustion.

The branch structure of Figure 3 is visible here as geometry. Configurations that start in the same place separate according to whether regulation holds: those keeping $R \gtrsim 0.5$ run to the right along the strip into the advanced regimes, while stage 4a drops to the lower boundary and trips the collapse proxy. [HYPOTHETICAL]

7.5 Decomposing the silence

Finally, Eq. (3) lets us ask which factor a null result actually constrains. Take stage 4c, with $O = 0.037$, $\kappa = 1$ and $P_{\text{search}} = 0.5$. The detection probability for a single civilization is

$$P_{\text{det}} = (1 - e^{-\kappa O})P_{\text{search}} = 0.018, \quad (18)$$

against $P_{\text{det}} = 0.50$ for the expansionist stage 4b at $O = 27$. So the same underlying population is about 27 times less likely to be detected in the quiet regime than in the loud one — with no change in N_{true} whatsoever, and nobody going extinct. [HYPOTHETICAL]

We want to be careful about what this does not show. It does not show that civilizations are quiet, and it certainly does not show that quietness explains the Great Silence. What it shows is narrower and, we think, more useful: under this framework a null result is compatible with a wide range of N_{true} , so anyone attributing non-detection to rarity is making an argument about O and P_{search} whether they say so or not. Usually they do not say so.

8 Discussion

8.1 Relation to existing solution classes

This framework is not a rival to the standard families of Fermi-paradox explanations. It is a way of writing several of them in a common state space, where they can be compared instead of merely asserted at one another.

Great Filter. A filter is a statement about where probability mass is lost. Here, a filter ahead of us shows up as a region of parameter space from which trajectories reach the collapse proxy. The model also makes something explicit that is usually left implicit: a filter acting on *observability* rather than on survival is equally possible, and using non-detection alone you could never tell the two apart (Hanson, 1998; Ćirković, 2018). [HYPOTHETICAL]

Grabby aliens. Hanson et al. (2021) argue that if loud, expanding civilizations explain human earliness, quiet civilizations have to be rare too. This is the sharpest challenge to any low-observability account, ours included, and we do not claim to answer it. What the framework contributes is somewhere to state the disagreement precisely: the grabby-aliens argument constrains the *joint* distribution of expansion behaviour and timing, which in our terms is a constraint on the population distribution of the expansion term X and on the branch probabilities at stage 3 — and both of those are quantities this model takes as given rather than derives. Coupling the two is the obvious next piece of work.

Zoo and non-interference hypotheses. These correspond to a stealth term S that is large and deliberate (Ball, 1973; Bracewell, 1960). The framework accommodates them comfortably, but it cannot tell deliberate quietness from emergent efficiency, because both enter Σ in exactly the same way. That degeneracy is worth stating rather than glossing: behavioural and thermodynamic explanations of quietness are not separable by observability alone, and no amount of better instrumentation changes that.

Percolation and expansion limits. Models in which settlement stalls (Landis, 1998; Forgan, 2009; Armstrong and Sandberg, 2013) act on X , the expansion contribution to production, and are compatible with the “inward” variant in Table 2.

8.2 Thermodynamic discipline

The floor O_{floor} is not a numerical convenience. Any civilization that uses energy radiates, and true invisibility would need assumptions about physics we have no basis for making (Dyson, 1960; Kardashev, 1964).

Putting the floor into the equation rather than into the prose buys something concrete: it makes the strong claim unavailable. You cannot accidentally conclude from this model that a civilization vanished. You can only conclude that it became hard to detect at a stated sensitivity κ and a stated coverage P_{search} . [ESTABLISHED]

That is exactly the difference between the corrected and uncorrected formulations of §7.2. The uncorrected version permitted arbitrarily negative observability, and so permitted — and in fact produced, and cheerfully labelled — a conclusion its own thermodynamic commitments forbid.

8.3 Implications for search strategy

If the low-observability regime is populated at all, it carries a testable consequence. In that regime, production is dominated by residual energy use rather than by broadcast or expansion, because the directed and efficient channels are precisely the ones suppression acts on hardest. Search programmes sensitive to waste heat and infrared excess would then keep their sensitivity where narrowband beacon searches lose theirs (Dyson, 1960; Wright et al., 2014; Garrett, 2015).

This is a conditional, not a recommendation. It holds if the regime is populated, and that claim is labelled [HYPOTHETICAL]. Its value is that it is the kind of statement that could turn out to be wrong: sustained null results from infrared surveys at improving sensitivity would push against it (Griffith et al., 2015; Wright et al., 2022).

Table 6: Claim-strength summary. [ESTABLISHED]: supported by observation or standard theory. [PLAUSIBLE]: compatible with known theory. [HYPOTHETICAL]: a model assumption or model output. [SPECULATIVE]: weakly constrained.

Claim	Strength
Existence, detectability and observation are distinct; non-observation constrains their product	[ESTABLISHED]
Planets are common; astrophysical Drake factors are comparatively well constrained	[ESTABLISHED]
Search coverage P_{search} is small and must appear explicitly in any inference	[ESTABLISHED]
Energy use implies a non-zero signature floor; true invisibility is unavailable	[ESTABLISHED]
Regulation confined to $[0, 1]$ and observability bounded below (Props. 1, 2)	[ESTABLISHED]
Peak-then-decline follows when suppression outpaces production	[PLAUSIBLE]
Detectability depends on civilizational behaviour and technology, not only on existence	[PLAUSIBLE]
A high-capability, low-observability regime is reachable under plausible parameters	[HYPOTHETICAL]
Quiet regimes favour waste-heat searches over narrowband beacon searches	[HYPOTHETICAL]
The Lambert boundary is a useful auxiliary diagnostic	[HYPOTHETICAL]
Recursive capability transitions change observability trajectories	[HYPOTHETICAL]
Quantum computation materially accelerates recursive self-improvement	[SPECULATIVE]
All stable high-capability civilizations become quiet	[SPECULATIVE]

8.4 On the role of recursive and quantum capability

The model treats A_{rec} as an exogenous driver and does not pretend to explain where recursive capability comes from. Quantum computation enters only as one possible contributor to that driver — a way the simulation, search and optimisation problems gating technological progress might be accelerated (Shor, 1994; Preskill, 2018).

We attach no magnitude to this, and there is a more interesting point to make than a caveat: its sign of effect on *observability* is not determined at all. Faster capability growth raises production through E and I , but it also raises suppression through C and S , and Eq. (9) says only the ratio matters. A civilization that computes far more efficiently might end up louder or quieter, depending entirely on which term scales faster. Anyone asserting a direction here is asserting a scaling law they have not written down. [SPECULATIVE]

8.5 Claim strengths

Table 6 collects every substantive claim with its label. In a subject where the distance between a model result and a statement about the universe is so easily lost, we regard this as a necessary discipline rather than a decorative one.

9 Limitations and failure criteria

9.1 Limitations

No calibration. Not one coefficient in Eqs. (5)–(8) is constrained by data, because no relevant data exist. The model is dimensionless and exploratory. It supports statements of the form “if these mechanisms operate with these relative strengths, then this follows”, and it supports

nothing else. The numbers in §7 illustrate qualitative regimes. They are not estimates of anything, and should never be quoted as though they were.

Collapse is a proxy, not a trajectory. There is no mortality term. Stage 4a therefore raises the collapse criteria — regulation below $R_{\min}$, amplification exceeding regulation by more than Θ — while capability carries on quite happily. The “transient spike, then nothing” signature people associate with civilizational collapse is consequently *not* reproduced dynamically. Coupling mortality to regulatory exhaustion is the obvious fix and we have not done it.

Stiffness. Making suppression fractional costs something numerically. The relaxation rate Σ scales as $I^2 R$ under the superlinear variants, and an explicit method needs step sizes of order $1/\Sigma$, so integration cost grows linearly with the suppression weights. That bounds how far into the strongly-suppressing corner of parameter space we can economically go. An implicit or stiffness-switching integrator would remove the limit.

The Lambert boundary is auxiliary. As §5 establishes, C_R does not appear in the integrated system. Nothing in this paper depends on the Lambert boundary, and it must not be read as a bifurcation of the dynamics.

Quietness is degenerate. Deliberate stealth and emergent efficiency enter Σ in exactly the same way, so observability alone cannot tell them apart. A civilization hiding on purpose and one that simply got very good at compression look identical from here.

No feedback from capability to regulation. Equation (6) makes regulatory capacity depend on the exogenous drivers and on erosion by A_{rec} , but not on I at all (Figure 4). This is a real simplification and probably the most consequential one. A highly capable civilization would plausibly be better at building governance as well as better at eroding it, and which of those wins is not obvious. Closing the loop — say by making the building term depend on $A_{\text{ref}} I$ — is straightforward, and it would open up genuine bistability between well-regulated and poorly-regulated high-capability states. We think that is the most interesting thing anyone could do with this model next.

One civilization, not a population. The model integrates a single trajectory. Any statement about N_{obs} therefore assumes a population whose members all share parameters, which is plainly false. A population-level treatment would integrate over distributions of the drivers and the branch probabilities, and it is required before this framework can be compared quantitatively with grabby-aliens arguments (Hanson et al., 2021; Prantzos, 2020).

9.2 Failure criteria

A model of this kind is hard to argue with unless its author says in advance what would count against it. So:

1. **Non-robustness.** If the low-observability regime arises only for a narrow band of suppression scalings, or only under one choice from Table 2, then the conclusion is an artefact of that choice. §7 reports the dependence explicitly; the regime does not arise under observability model A.
2. **Assumed rather than derived decline.** If peak-then-decline could be obtained only by selecting a peaked $O(I)$, this framework would add nothing to the low-observability arguments that already exist. Equation (11) and §7.2 are our answer. Show that the

required scaling separation — Σ growing faster in I than Π — is physically implausible, and the answer fails.

3. **Collapse and quietness not separated.** If the model could not distinguish a civilization that ended from one that went quiet, it would reproduce the exact ambiguity it was built to dissolve. The regime criteria do separate them, but only as far as the collapse proxy means anything — see above.
4. **Thermodynamic violation.** Any result requiring $O \leq 0$ invalidates the physical interpretation outright. Proposition 2 makes that unreachable by construction, and the property-based tests of §6 check it empirically over 300 randomised configurations.
5. **Search coverage ignored.** Drop P_{search} and non-detection gets misread as evidence about N_{true} . It appears explicitly in Eq. (3) and must stay there.
6. **Empirical pressure.** Sustained null results from infrared and waste-heat surveys at improving sensitivity would argue against the quiet regime being populated, since that is precisely the channel it predicts stays open (Griffith et al., 2015; Garrett, 2015; Wright et al., 2022).

Only the last of those is empirical rather than internal. That asymmetry is a limitation of the subject rather than of the model, and we would rather name it than dress it up.

10 Conclusions

The Fermi paradox is usually modelled by varying how many civilizations arise. We have argued it should also be modelled by varying how long they stay externally observable, and that observability belongs in the state vector with its own equation of motion rather than in a coefficient.

Three results follow.

First, once existence, signal production and search coverage are pulled apart in Eq. (3), non-observation constrains their product and not N_{true} alone. That is a logical point rather than a discovery, but it is elided routinely, and it determines what a model of the paradox should vary in the first place. [ESTABLISHED]

Second, an observability equation has to be built so that it cannot leave the physical range. Writing suppression as a fractional rate acting on the emitted signal, above an explicit thermodynamic floor, achieves that by construction (Proposition 2), and treating erosion the same way confines regulatory capacity to the unit interval (Proposition 1). These are not cosmetic repairs. The uncorrected formulation produced — and confidently labelled “optimized low-observable” — a quiet civilization whose observability was -1431 .

Third, and this is the modelling result we would defend hardest, peak-then-decline observability does not have to be assumed. Because the quasi-steady observability is a ratio of production to suppression, the decline appears on its own whenever suppression scales faster in capability than production (Eq. (11)). A low-observability account of the Great Silence built this way does not assume its own conclusion, and it can be attacked where an argument should be attacked: on whether the scaling separation is plausible, rather than on the choice of a convenient curve. [PLAUSIBLE]

We have also reported a failure. The Lambert W threshold that named and motivated this work is not a property of the integrated system at all. The natural balance cancels I and leaves a logarithm, and recovering Lambert structure needs an auxiliary balance we have to declare rather than derive (§5). We keep the boundary as a diagnostic and depend on it nowhere.

Finally, this framework is verified rather than merely implemented. Closed-form solutions, an independent high-order integrator, quadrature of the observability equation, and property-based

testing over 300 randomised parameter sets together establish that the equations described here are the equations being solved. They establish nothing about whether those equations describe a real civilization, and we have worked to keep verification and validation from being confused.

The framework does not solve the Fermi paradox and was never meant to. Its purpose is to make a class of arguments about cosmic silence precise enough to be *wrong*: to say which assumptions produce which regimes, which conclusions survive a change of functional form, and what observation would count against them. The most useful next steps are a population-level treatment that can be set against grabby-aliens constraints, a collapse term that produces trajectories instead of flags, and a nondimensionalisation that cuts the parameter count down to the few groups that actually govern the behaviour.

Data and code availability

The simulator runs in a browser, with no installation and nothing to configure, at https://rehmozayub.github.io/Ulm_QH_recursive_observability_explorer/. Every preset discussed in §7 is reachable from its menu, and every parameter in Appendix C is exposed as a slider, so any claim made here can be pushed until it fails.

Source, verification harness and the script that regenerates every data figure are at https://github.com/RehmozAyub/Ulm_QH_recursive_observability_explorer. *[Mint a Zenodo DOI before submission and cite it here.]*

Acknowledgements

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A Notation

Every symbol used in the paper, in the order the variables, drivers, coefficients and derived quantities are introduced.

Symbol	Type	Meaning
τ	—	dimensionless time
I	state	effective capability
R	state	regulatory quality index, $R \in [0, 1]$
O	state	external observability, $O \geq O_{\text{floor}}$
A	driver	ordinary amplification pressure
A_{rec}	driver	recursive amplification
A_{ref}	driver	reflective amplification
Q	driver	institutional quality
a, b, c, s_I	coeff.	ordinary growth, recursive growth, regulatory damping, saturation
u, v, w, s_R	coeff.	reflective learning, institutional gain, erosion, decay
p, q, r	coeff.	energy-use, broadcast and expansion weights
m, n	coeff.	compression and stealth suppression <i>rates</i>
$F(I)$	function	recursion kernel
$\Pi(I)$	function	total signature production, $pE + qB_c + rX$
$\Sigma(I, R)$	function	total fractional suppression rate, $mC + nS$
$h(O)$	function	detection map, $1 - e^{-\kappa O}$
B, W	derived	regulation building and erosion aggregates, Eq. (6)
R_∞	derived	regulation equilibrium, $B/(B + W)$
O^*	derived	quasi-steady observability, Eq. (9)
C_R	derived	auxiliary regulation capacity, Eq. (16)
I_{crit}	derived	auxiliary Lambert boundary, Eq. (17)
$N_{\text{true}}, N_{\text{detectable}}, N_{\text{obs}}$	counts	existing, detectable, observed civilizations
$P_{\text{surv}}, P_{\text{det}}, P_{\text{search}}$	prob.	survival, detection, search coverage
κ	param	detection sensitivity
$O_{\text{floor}}, O_{\text{detectable}}$	param	thermodynamic floor; detection threshold
$I_{\text{advanced}}, R_{\text{min}}, \Theta$	param	capability, regulation and runaway thresholds
$\eta, \mathcal{E}$	param	regulation exponent; alignment factor (Lambert case)
$W(\cdot)$	function	Lambert W , principal branch

B Closed-form solutions

Two exact solutions exist and are used as verification oracles in §6.

B.1 Regulation

With constant drivers, Eq. (6) is linear in R and independent of I and O :

$$\frac{dR}{d\tau} = B - (B + W) R, \quad B = uA_{\text{ref}} + vQ, \quad W = wA_{\text{rec}} + s_R, \quad (19)$$

so that

$$R(\tau) = R_\infty + (R_0 - R_\infty) e^{-(B+W)\tau}, \quad R_\infty = \frac{B}{B + W}. \quad (20)$$

Because $B, W \geq 0$ we have $R_\infty \in [0, 1]$ immediately, which is the substance of Proposition 1. Equation (20) also gives the relaxation time $1/(B + W)$: regulation responds faster when either building or erosion is strong.

B.2 Capability in the reduced stage models

For stages 0–2 the regulatory damping term is inactive and recursive amplification is absent, so Eq. (5) reduces to the logistic equation

$$\frac{dI}{d\tau} = aA I - s_I I^2, \quad (21)$$

with solution

$$I(\tau) = \frac{K_I}{1 + \left(\frac{K_I}{I_0} - 1\right) e^{-aA\tau}}, \quad K_I = \frac{aA}{s_I}. \quad (22)$$

With the stage-1 parameters ($a = 0.30$, $A = 1$, $s_I = 0.10$, $\lambda = 0.30$) this gives $K_I = 3.0$ exactly, and hence a limiting observability under model A of

$$O_\infty = 1 - e^{-\lambda K_I} = 1 - e^{-0.9} = 0.5934. \quad (23)$$

Both are reproduced by the solver to four decimal places (Table 5), which is a stringent check because it tests the integrator, the parameter plumbing and the stage configuration simultaneously.

B.3 The general Bernoulli form

More generally, whenever recursive amplification is absent ($bA_{\text{rec}} = 0$), Eq. (5) is a Bernoulli equation $dI/d\tau = g(\tau)I - s_I I^2$ with $g(\tau) = aA - cR(\tau)$. The substitution $z = 1/I$ linearises it to $z' + gz = s_I$, giving

$$I(\tau) = \left[e^{-G(\tau)} \left(\frac{1}{I_0} + s_I \int_0^\tau e^{G(s)} ds \right) \right]^{-1}, \quad G(\tau) = \int_0^\tau g(s) ds, \quad (24)$$

where G is available in closed form because R is (Eq. (20)). This provides a quadrature-based check independent of the ODE solver.

C Parameters

Table 7 gives the default values and admissible ranges used throughout. All quantities are dimensionless. None is calibrated against data (§9); the ranges bound exploration rather than expressing prior belief.

Stage configurations differ from these defaults only in the drivers, the functional selections of Table 2, and the initial conditions. Stages 0–2 additionally disable the regulatory damping term, reproducing the reduced models of the source framework literally rather than approximating them by parameter choice.

D Verification protocol

D.1 Configuration matrix

Every oracle in §6 is evaluated over the same twenty configurations: the eight stage presets; five configurations exercising each recursion kernel of Table 2; four exercising each static observability model; one with time-varying drivers; one Lambert configuration; and one that selects the non-default variant of every observability component. The intent is coverage of the registry rather than of parameter space, which is what the randomised testing of oracle E is for.

Table 7: Default parameter values and admissible ranges.

Symbol	Default	Range	Role
<i>Capability, Eq. (5)</i>			
a	0.30	$[0, 2]$	ordinary growth
b	0.20	$[0, 2]$	recursive growth
c	0.50	$[0, 3]$	regulatory damping
s_I	0.10	$[0, 1]$	capability saturation
<i>Regulation, Eq. (6)</i>			
u	0.40	$[0, 2]$	reflective learning
v	0.30	$[0, 2]$	institutional gain
w	0.50	$[0, 3]$	erosion by recursive amplification
s_R	0.15	$[0, 1]$	regulation decay
<i>Observability, Eq. (8)</i>			
p	0.50	$[0, 3]$	energy-use weight
q	0.30	$[0, 3]$	broadcast weight
r	0.40	$[0, 3]$	expansion weight
m	0.50	$[0, 3]$	compression suppression rate
n	0.30	$[0, 3]$	stealth suppression rate
<i>Drivers</i>			
$A, A_{\text{rec}}, A_{\text{ref}}, Q$	1.0	$[0, 5]$	amplification and institutional drivers
<i>Shape and detection</i>			
K	5.0	$[0.1, 50]$	half-saturation of the saturating kernel
λ	0.30	$[0.01, 3]$	observability scale
κ	1.0	$[0.01, 10]$	detection sensitivity
P_{search}	0.50	$[0, 1]$	search coverage
P_{surv}	0.80	$[0, 1]$	survival probability
<i>Thresholds</i>			
Θ	5.0	$[0.1, 50]$	runaway ratio A_{rec}/R
I_{runaway}	50	$[1, 1000]$	runaway capability
R_{min}	0.05	$[0, 1]$	regulatory exhaustion
I_{advanced}	5.0	$[0, 100]$	advanced capability
$O_{\text{detectable}}$	0.10	$[0, 1]$	detection threshold
O_{floor}	10^{-3}	$[0, 0.1]$	thermodynamic floor
<i>Numerics</i>			
τ_{max}	50	$[1, 500]$	integration horizon
I_{abort}	10^4	—	runaway guard (§6)
rtol / atol	10^{-6} / 10^{-9}	—	solver tolerances

D.2 Property-based testing

Oracle E draws parameters uniformly from the ranges in Table 7, together with a randomly chosen recursion kernel and randomly chosen suppression forms, and asserts

$$O(\tau) \geq 0 \quad \text{and} \quad 0 \leq R(\tau) \leq 1 \quad \text{for all sampled } \tau. \quad (25)$$

Three hundred draws produced no violations, with global extrema $\min O = 2.16 \times 10^{-3}$ and $R \in [4.0 \times 10^{-3}, 0.999]$.

This is stronger evidence than checking the presets, and the reason is worth spelling out: the propositions assert *structural* properties, so a bound that held only for tuned configurations would not be a bound at all. It would be a property of our taste in parameters.

The randomised search restricts the recursion kernel to the non-explosive variants. The blow-up kernels terminate on the runaway guard, so including them would test the guard rather than the invariant.

D.3 Reproduction

```
# Test suite: 62 tests
cd recursive_observability_explorer
PYTHONPATH=. python -m pytest tests -q

# Regenerate every data figure in this paper
cd paper
python make_figures.py

# Build the paper (latexmk needs Perl, which a stock MiKTeX lacks)
pdflatex main.tex && bibtex main && pdflatex main.tex && pdflatex main.tex
```

Environment used: Python 3.11 with NumPy 2.4.2 and SciPy 1.17.1 (Harris et al., 2020; Virtanen et al., 2020). Figures 1–4 are TikZ and need no external data.

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